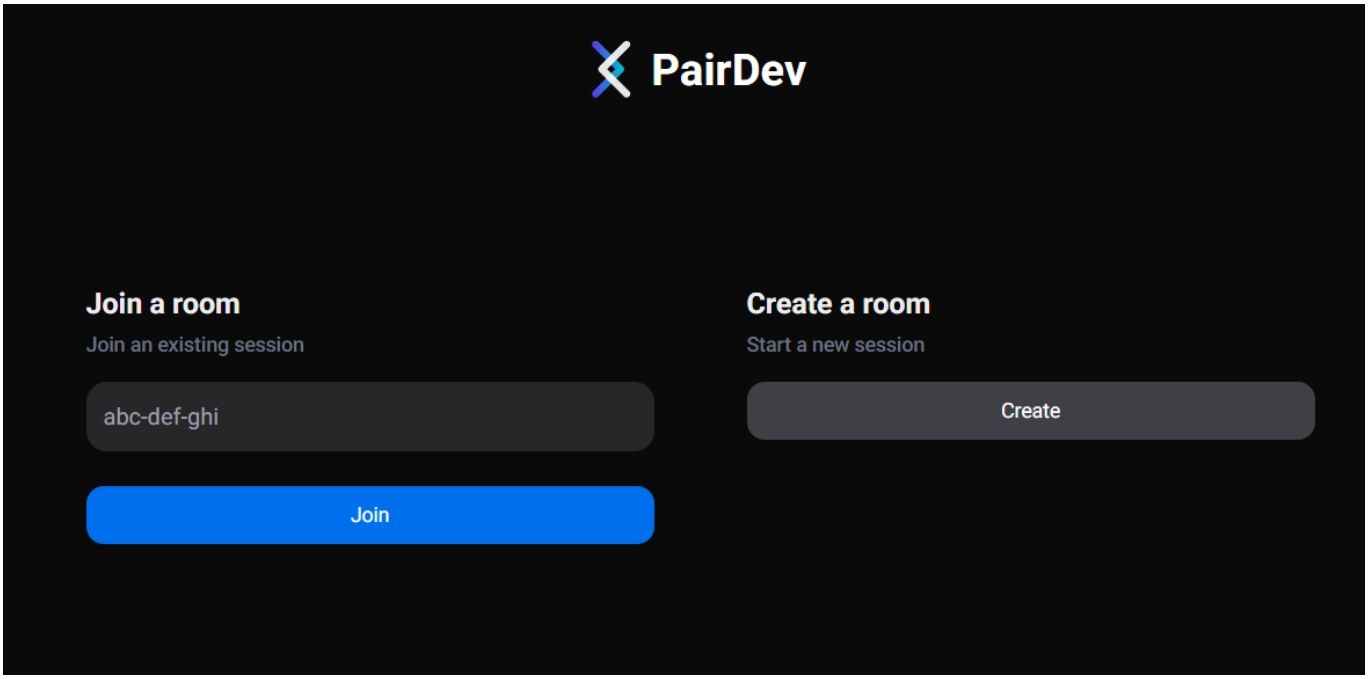
PairDev Manual

PairDev is a browser-based remote pair-programming tool for two people. It combines a shared code editor, driver and navigator roles, a shared terminal, a collaborative whiteboard, code annotation, GitHub import for public repositories, and session analytics.

1. Start a session

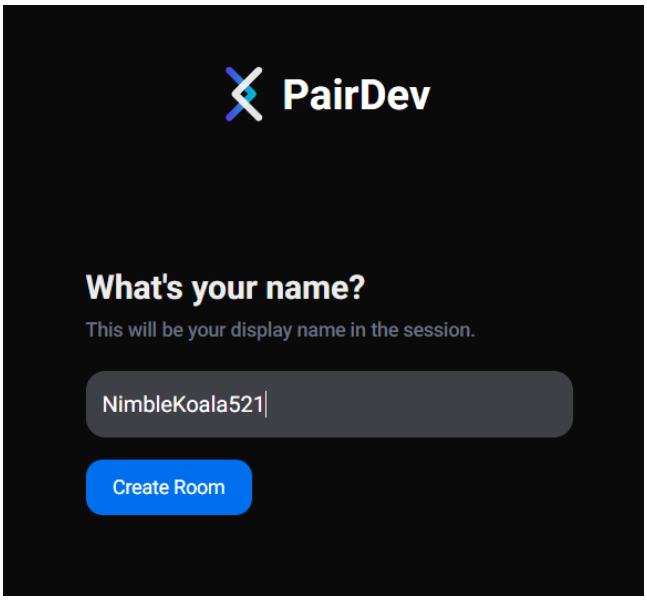
There are two ways to begin using PairDev:

- **Create a room** to start a new collaborative session.
- **Join a room** using a room code shared by another user.



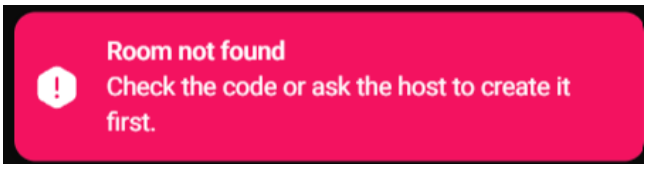
After selecting or creating a room, enter your display name.

- This name is shown to the other participant.
- It helps identify you in the people list, role controls, and live collaboration features.



Notes

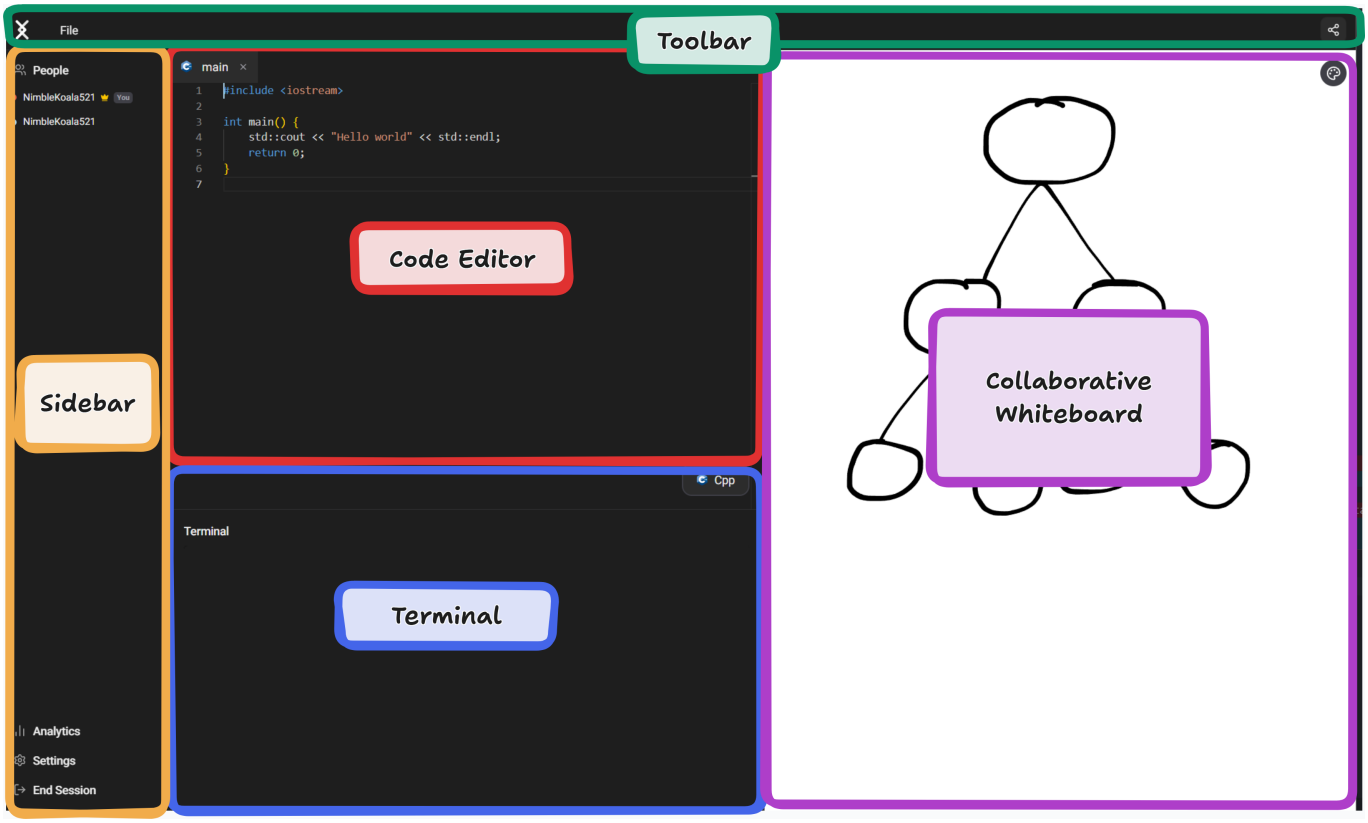
- Room codes should follow the format `xxx-xxx-xxx`.
- PairDev shows an error toast if the room code format is invalid.
- PairDev also shows an error toast if the room does not exist yet.



2. Understand the workspace

The main workspace is divided into four core areas:

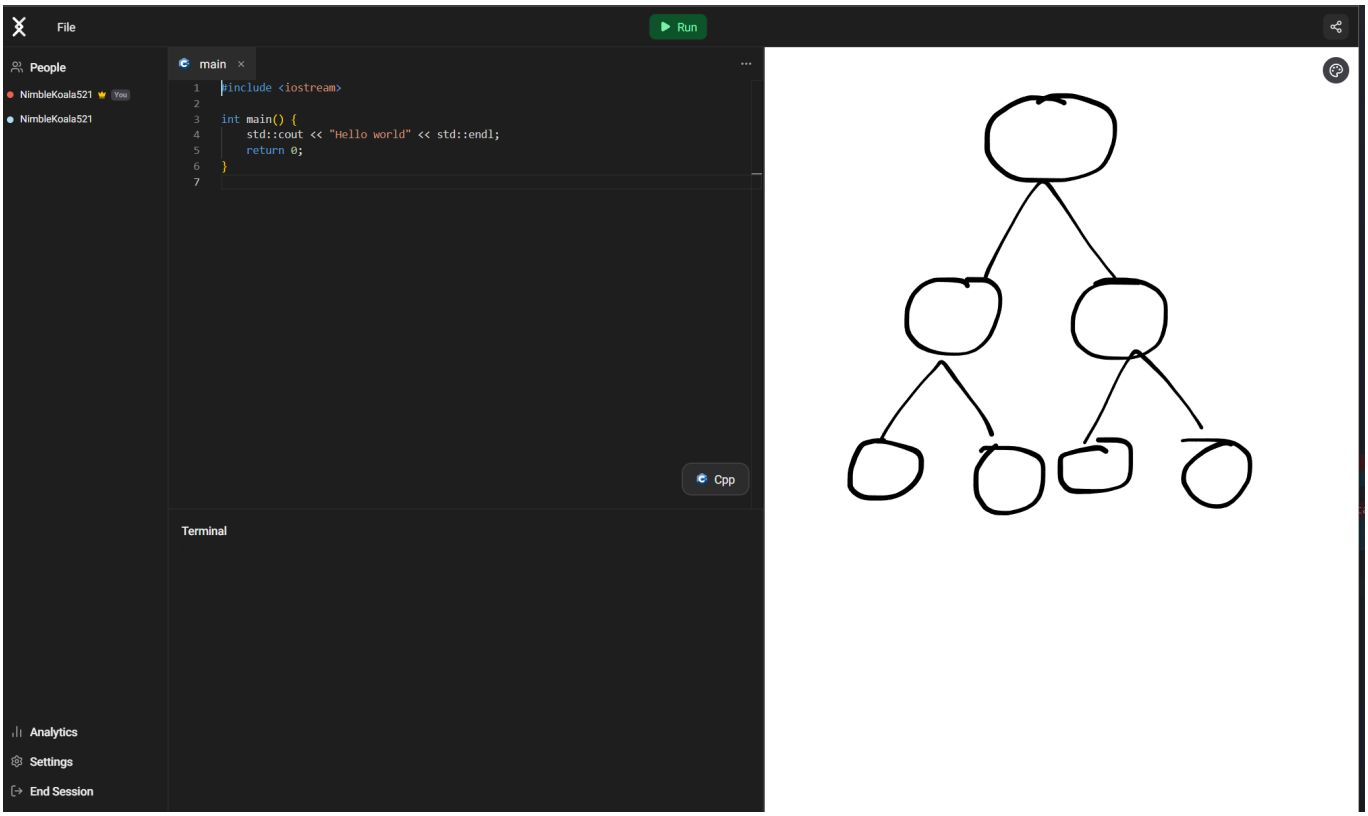
- **Sidebar**
  - shows participants
  - opens analytics
  - opens settings
  - ends the session
- **Code editor**
  - main programming area
  - shared between participants
- **Terminal**
  - shows execution output
  - shared between participants
- **Collaborative whiteboard**
  - used for diagrams, sketches, and visual explanation



What the top toolbar is used for

The toolbar gives access to important session actions, such as:

- running code
- sharing the invite link
- opening language-related controls

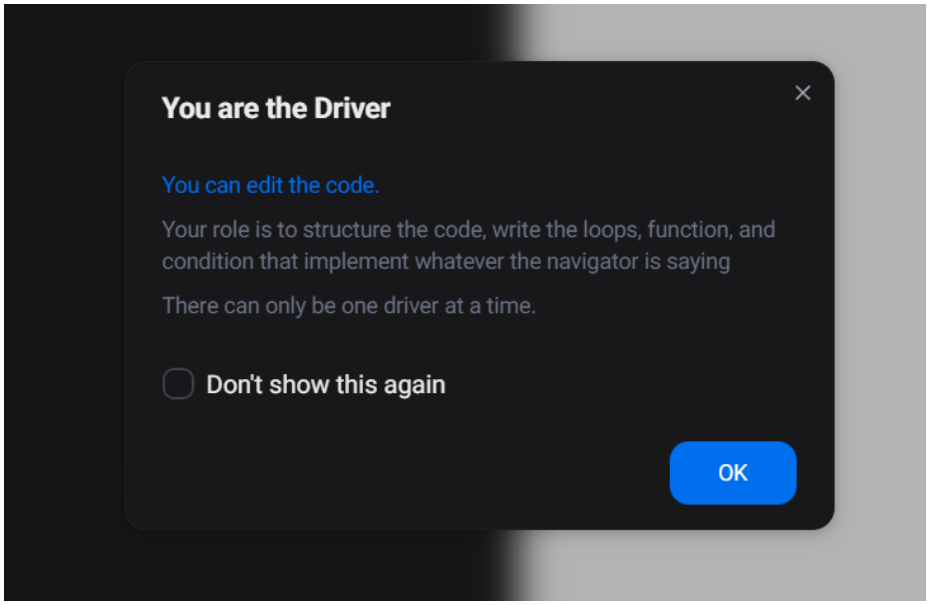


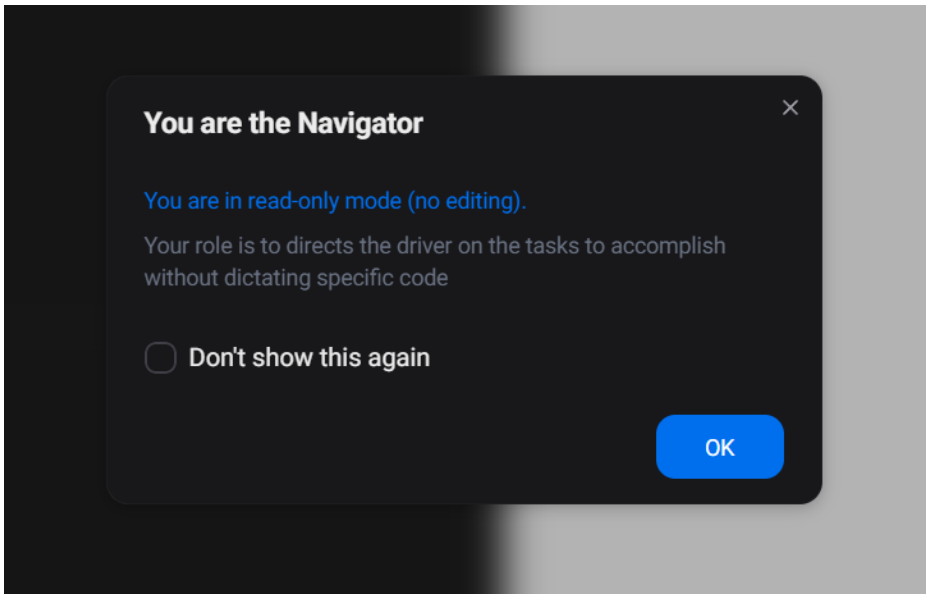
3. Understand roles

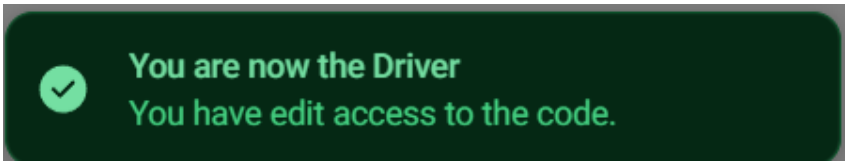
PairDev uses two main roles:

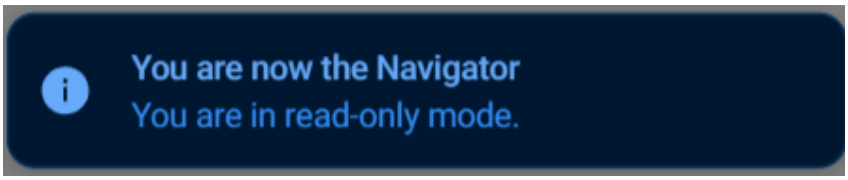
- **Driver**
  - can edit the code
  - is responsible for writing the code
- **Navigator**
  - cannot edit the code
  - reviews the work and guides the session

When your role changes, PairDev explains it through both a dialog and a toast.









Role rules

- Only the **room owner** can change roles.
- Only one participant can be the **driver** at a time.
- A navigator stays in **read-only** mode in the editor.

4. Collaborate in the editor

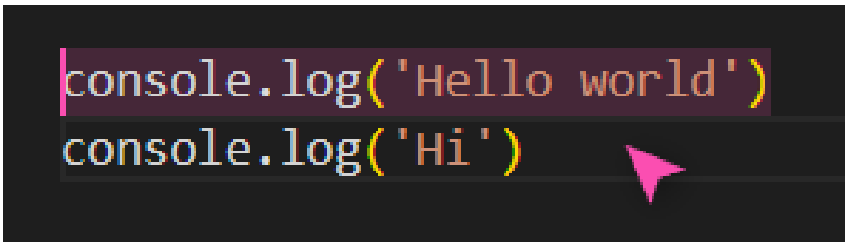
The code editor is the main work area.

- The **driver** writes and edits the code.
- The **navigator** sees updates in real time.
- Both participants can discuss the code while looking at the same shared document.

Live collaboration cues

PairDev also shows live presence information in the editor, such as:

- the collaborator's cursor label
- text highlight around the active area
- visual indication of where the other participant is working

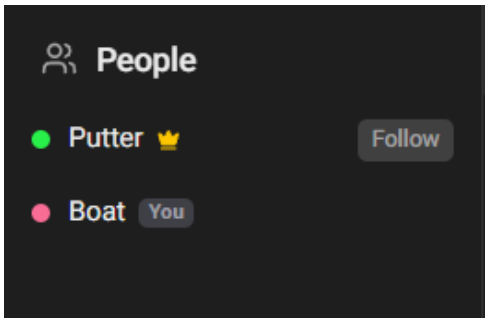


5. Use the participant list and follow mode

The people panel helps you understand who is in the room and who controls the session.

It shows:

- participant names
- the room owner
- your own identity
- whether follow mode is available



Follow mode

When you are the **navigator**, you can use **Follow** to keep your editor view aligned with the driver.

This is useful when you want to:

- stay focused on the same part of the code
- avoid manually searching for the driver's current location
- follow the flow of the session more easily

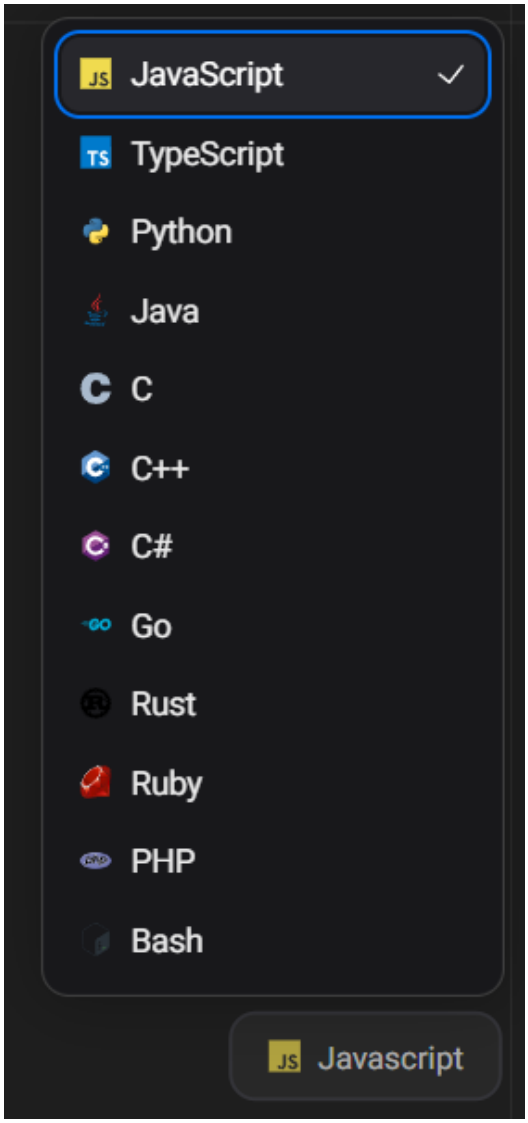
6. Select a language and run code

Before running code, choose the programming language from the language selector.

Supported languages include:

- JavaScript
- TypeScript
- Python
- Java
- C
- C++
- C#
- Go

- Rust
- Ruby
- PHP
- Bash



Then:

1. write code directly in the editor, or
2. import code from a public GitHub repository,
3. click **Run**.

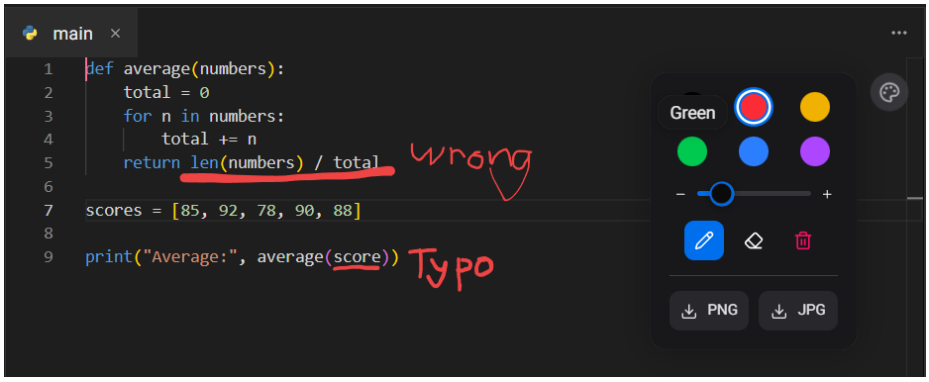
The result appears in the shared terminal for both participants.

7. Use code annotation

Code annotation lets you draw directly over the editor.

Use it to:

- point out mistakes
- highlight important lines
- mark suspicious logic
- explain code visually during discussion

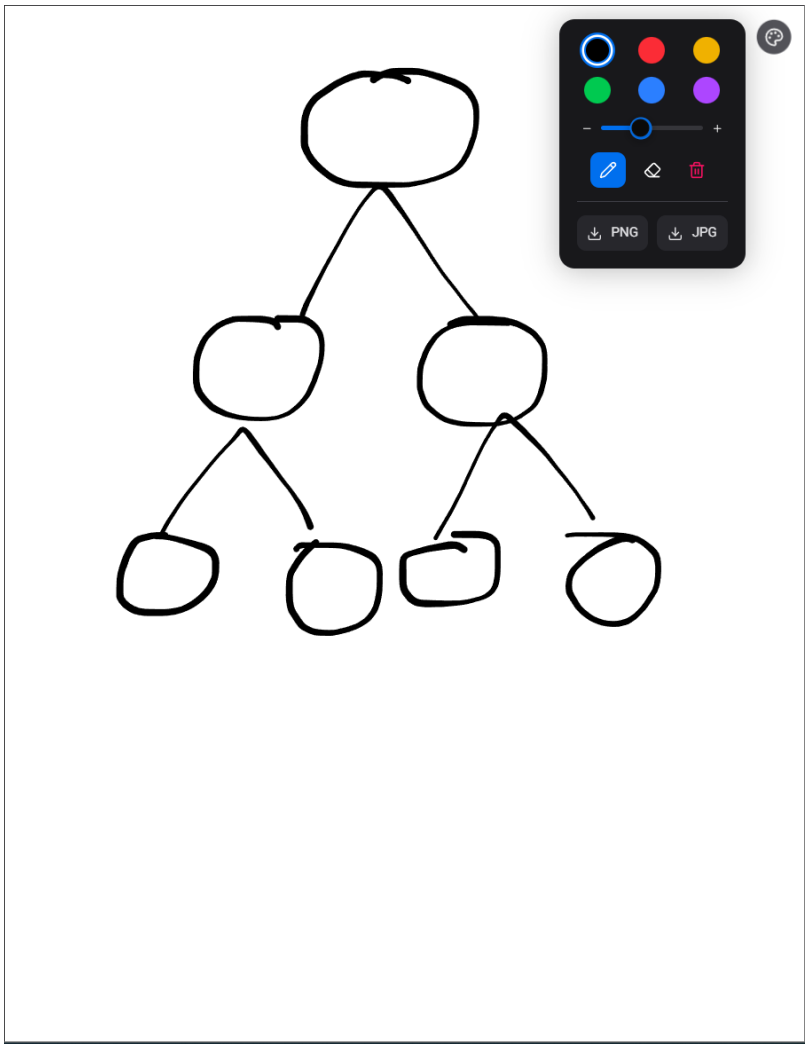


8. Use the collaborative whiteboard

The collaborative whiteboard is a separate shared space for drawing.

It is useful for:

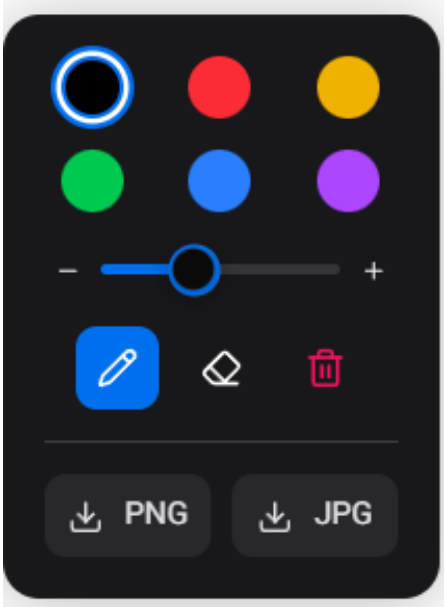
- tree structures
- diagrams
- sketches
- algorithm explanations
- general visual discussion that does not need to sit directly on top of the code



Whiteboard tools

The whiteboard and annotation tools include:

- colour selection
- brush size control
- pen tool
- eraser tool
- clear action
- export to PNG
- export to JPG



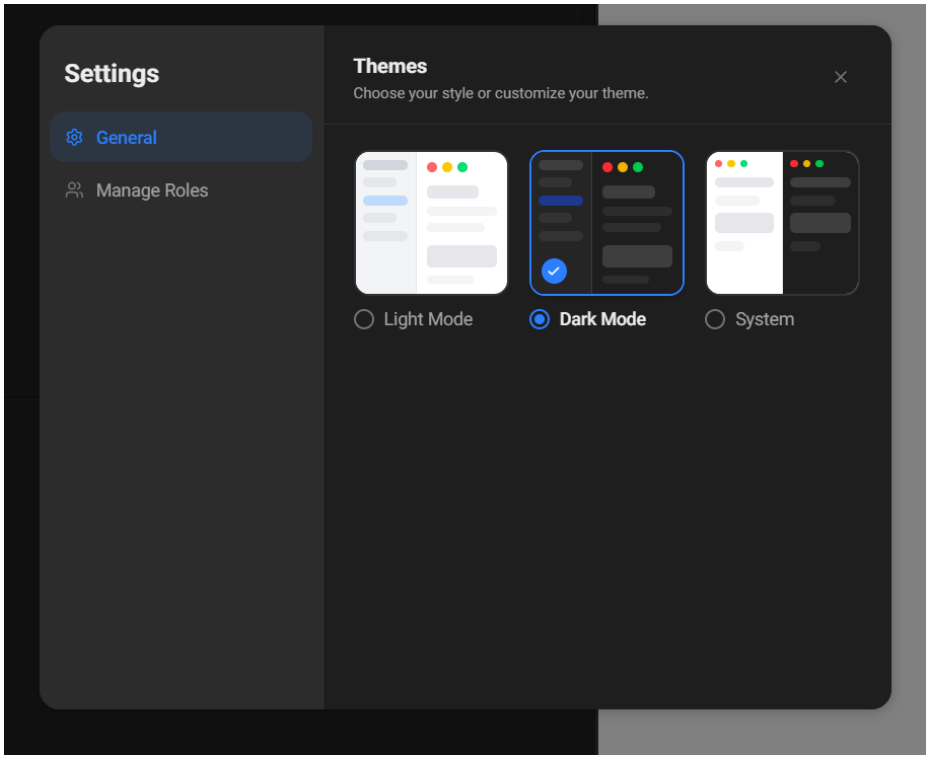
9. Open settings and manage roles

The settings modal provides both general preferences and room controls.

General settings

From the general settings tab, you can change the visual theme:

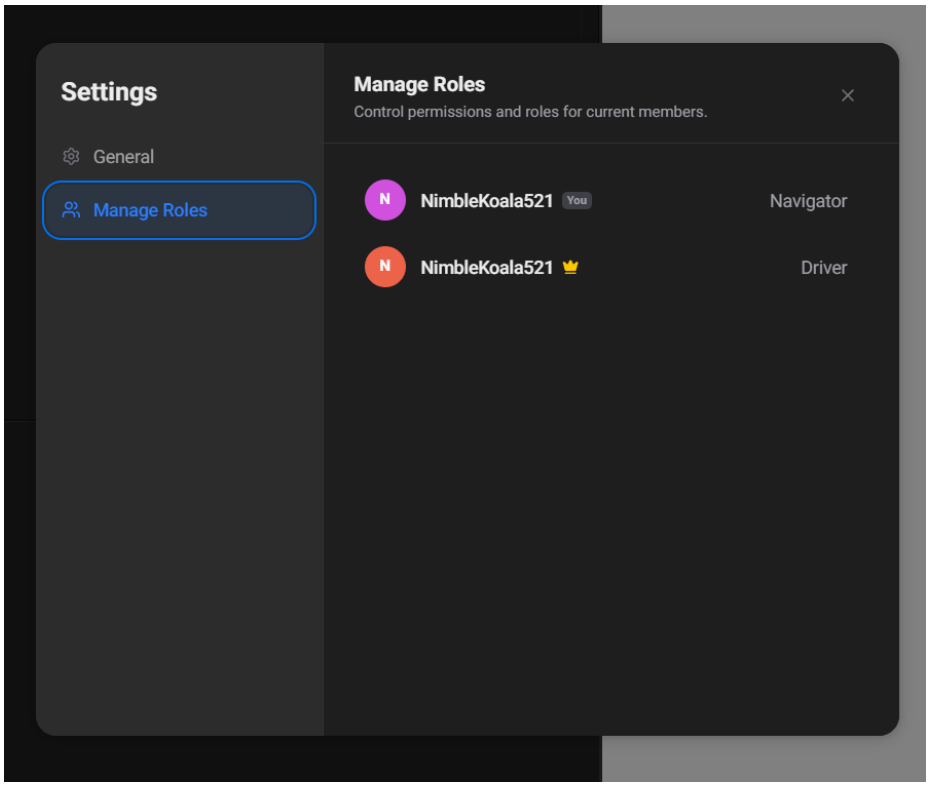
- **Light mode**
- **Dark mode**
- **System**



Manage roles

From the manage roles tab, the room owner can:

- inspect current participants
- see who is the driver or navigator
- update roles during the session

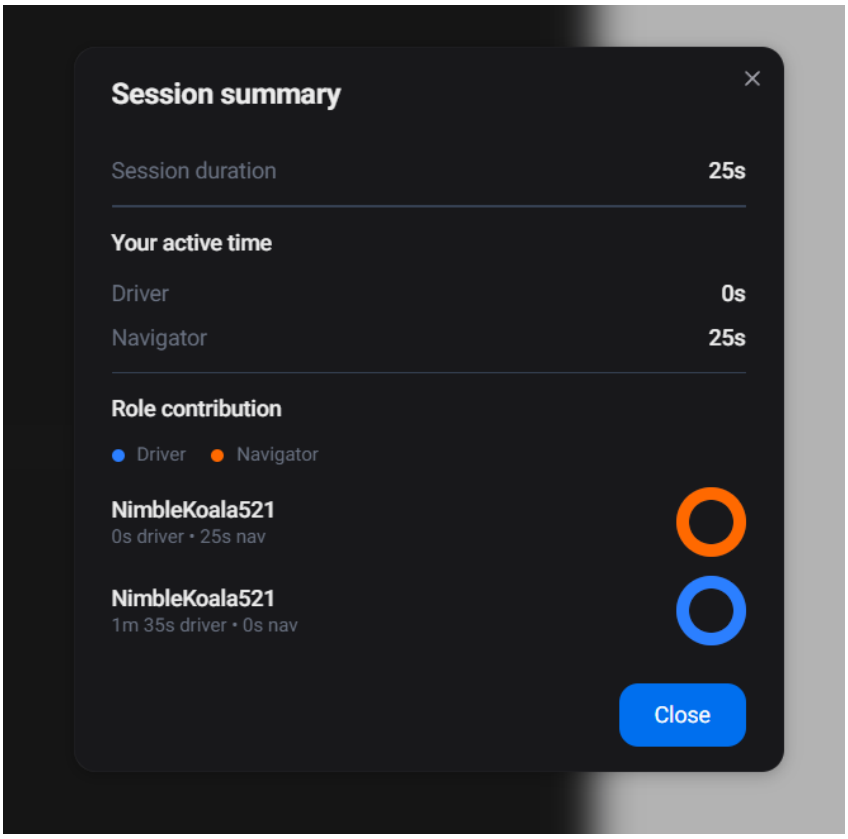


10. View analytics

PairDev provides a session summary after or during a session.

The analytics view shows:

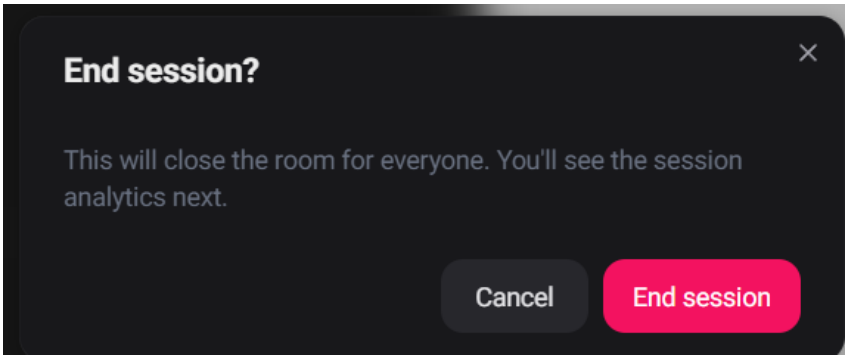
- total session duration
- your active time
- time spent as driver
- time spent as navigator
- role contribution for participants



11. End the session

When the session is finished, the owner can end it for everyone.

Before the room closes, PairDev shows a confirmation dialog so that the action is not triggered accidentally.



12. Quick tips

- Share the invite link after creating a room.
- Use the collaborative whiteboard for diagrams.
- Use the annotation tool for code-focused discussion.
- Use follow mode when the navigator wants to stay aligned with the driver.
- Check the participant list to confirm who is the owner and who is the driver.
- If GitHub import is used, only public repositories are supported.

13. Common issues

I cannot edit the code

Possible reason:

- You are currently the **navigator**.

What to do:

- Ask the owner to switch your role to **driver**.

The room does not open

Possible reasons:

- The room code format is invalid.
- The room does not exist yet.

What to do:

- Check the room code.
- Ask the host to create the room again.

GitHub import does not work

Possible reasons:

- The repository is private.
- The URL is invalid.

What to do:

- Make sure the repository is public.
- Check that the GitHub URL is correct.

I lost ownership after reopening

Possible reason:

- Ownership recovery depends on the current browser session.

What to do:

- Reopen the room in the same browser session if possible.
- Avoid clearing session data during an active session.